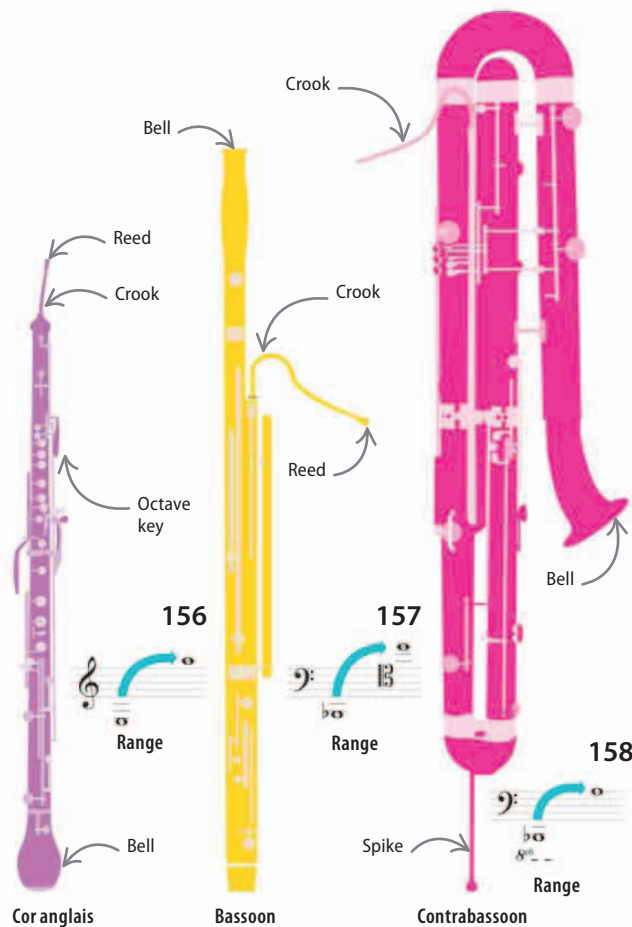
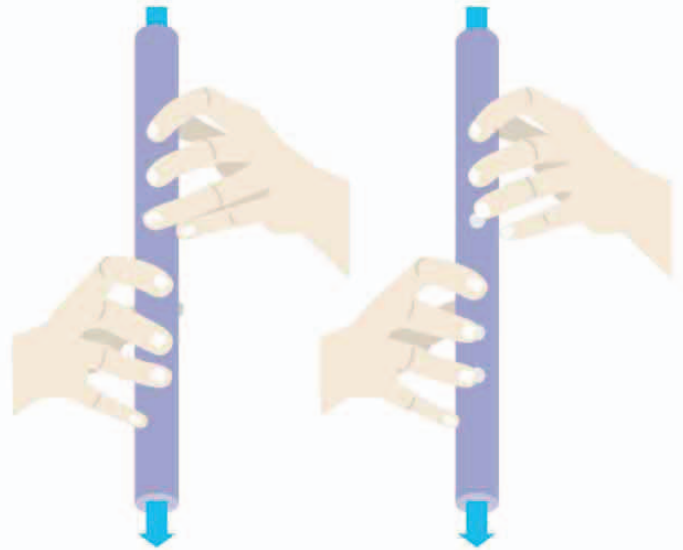




Producing the sound

When the column of air in a woodwind instrument's tube is made to vibrate, sound is produced. A short, narrow tube gives a higher pitch than a long, wide one. The holes in the tube allow the air to escape, effectively shortening the tube.



△ Changing the pitch

The player's fingers cover the holes in various combinations, creating different pitches. The holes beyond the reach of the player's fingers may be closed or opened by finger-operated keywork.

Reeds and mouthpieces

Woodwind players blow into a cane reed or a mouthpiece to make the column of air in the tube vibrate. This gives each woodwind instrument its distinctive tone color, or timbre. The sweet breathiness of the flute and clarinet contrast well with the penetrating, edgy quality of the double-reeds.

▷ Single reed

The air blown into the mouthpiece of a clarinet and saxophone causes a single piece of thin cane, the reed, to vibrate against the hard lower surface of the mouthpiece.



△ Flute mouth plate

The player directs air against the sharp metal edge of the mouth plate. This causes the column of air to vibrate, producing a sound like blowing across the top of a bottle.



Oboe reed

Bassoon reed

◁ Double reeds

Oboes and bassoons are played with double reeds, two lengths of cane bound together at the base. Air is blown at high pressure through a tiny aperture between the two blades of cane.